Equal Pay for Equal Work for Equal Pay

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Pay “fairness” is a big issue

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- But what does “fairness” mean?

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Black proverb: "you have to be twice as good to go half as far as a white person"– Violates Becker's outcome test!

Cherie Priest, Allegedly @cmpriest · 15h

tfw you don't just have to work twice as hard as a man, you can achieve double his success - only to have his quantifiably lesser accomplishments deemed "equally impressive"

Lila Byock @LByock · Jul 27

This headline is so bizarre.

IndieWire

You might like this · 21h ·

"Oppenheimer" could reach half those numbers, an equally impressive achievement.

indiewire.com

Margot Robbie Was Right: 'Barbie' Will Be a Billion-Dollar Movie

Two features about these fairness notions

1. At an *individual level*
   - Most of literature on equal pay practices studies group-level consequences, not individual-level
   - But if Ann must be paid same as Bill and Ann must be paid same as Charlie, then transitivity $\Rightarrow$ Bill must be paid same as Charlie.
   - So equal pay constraint applies to all pairs of workers, and e.g. firm cannot skirt law by paying men and women same on average but paying individuals differently

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2. *Complete information* standard of proof (i.e. firm cannot skirt law by saying it isn't sure if workers are equally productive or not)
   - Laws explicitly require firms to have a uni-dimensional measure of productivity

What we do

1. Theoretically: model effects/restrictions of pay "fairness"
   - Show various fairness notions boil down to a simple monotonicity condition
   - Show that this condition is restrictive. May prohibit wage bargaining, matching outside offers.

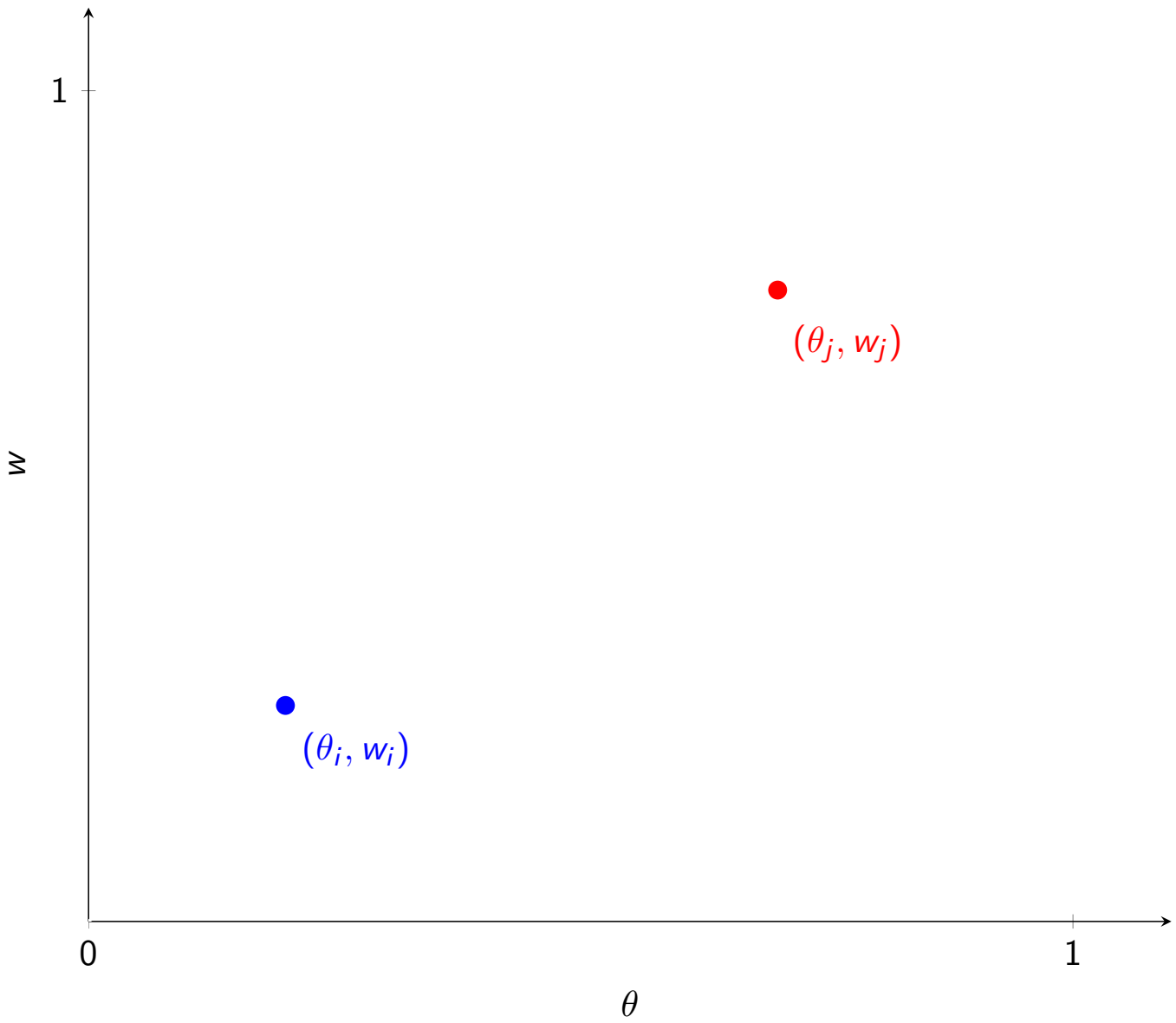
2. Empirically (earliest stages): do firms abide by monotonicity?
   - Study US academics, where wages publicly available for some universities and productivity (research) is observable
   - Further avenues: effect of job transition? Effect of new equal pay law enactment?

Model and fairness notions

Setup

- Single firm employing N employees $\{1, ..., N\}$
- Each worker i has productivity $\theta_i \in [0, 1]$ and wage $w_i \in [0, 1]$

Example with $N = 2$

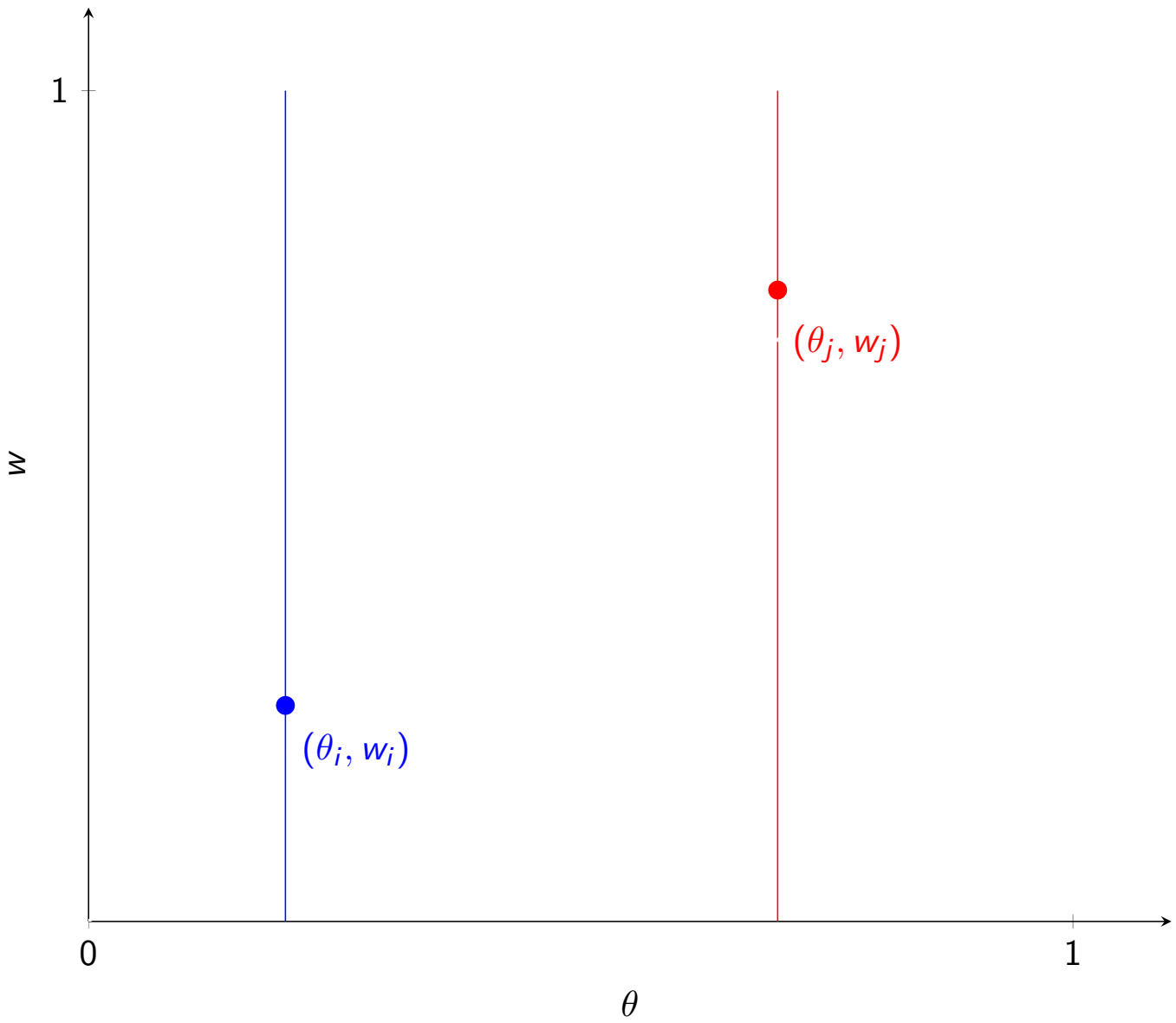


(weak) Equal Pay for Equal Work

Definition

The firm exhibits *weak Equal Pay for Equal Work (wEPEW)* if for any two workers i and j with $\theta_i = \theta_j$ it is the case that $w_i = w_j$.

wEPEW $\Rightarrow$ no one on solid lines



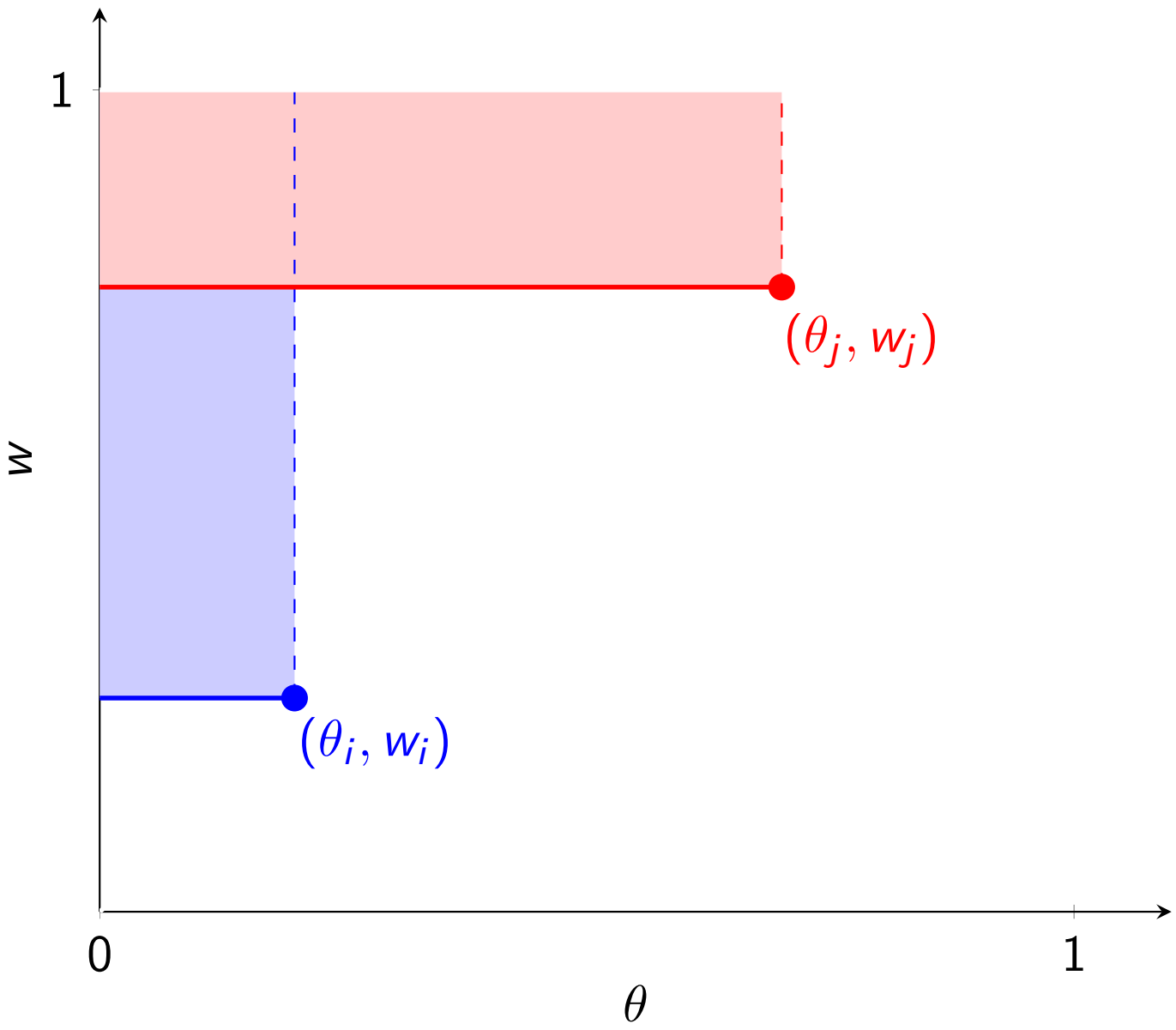
(weak) Equal Work for Equal Pay

Definition

The firm exhibits *weak Equal Work for Equal Pay (wEWEP)* if for any two workers i and j with $w_i > w_j$ it is the case that $\theta_i \geq \theta_j$.

- Continuous outcome version of Becker's outcome test
- Has an incentive basis also: Suppose there is a violation so $w_i > w_j$ but $\theta_i < \theta_j$. Then j could destroy her productivity until $\theta_j = \theta_i$ and then sue the firm for EPEW violation

wEWEP $\Rightarrow$ no one in shaded regions



Equal Pay for Equal Work

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2. if $\theta_i > \theta_j$, then $w_i \geq w_j$.

- "equal pay for equal or greater work"
- Case law suggests this may be the right interpretation of EPEW laws (see e.g. U.S. Equal Employment Opportunity Commission v. Aimbridge Hospitality, LLC and AH 2007 Management, LP Case No. 19-cv-914)

Putting it all together

Proposition

The following conditions are all equivalent:

1. wEPEW + wEWEP,
2. EPEW,
3. there exists a non-decreasing function $\psi : [0,1] \to [0,1]$ such that for each worker i of productivity θ, $w_i = \psi(\theta)$.

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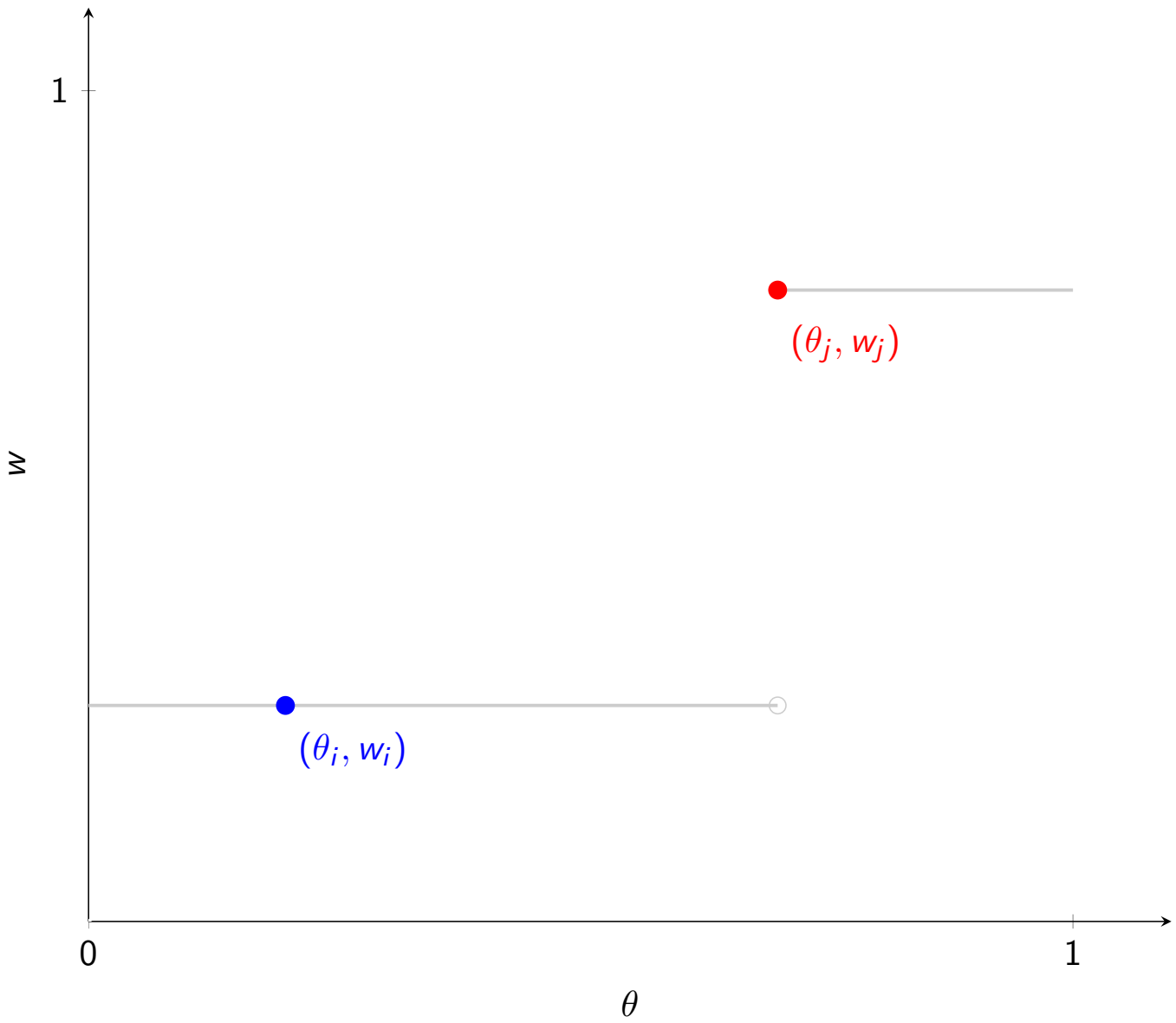
- Can also define a "strong" version of EWEP, which if placed in result above changes "non-decreasing" to "strictly increasing"

Monotonic wage schedule in gray



Aside: Other fairness notions?

- CEO should not make more than X times what workers make
- Equal pay for "similar" work
- Equal pay for all
- These all add upper bound to slope of wage function

Monotonicity restrictive, unintended consequences?

- RI Pay Equity Act: "An employer may, for the limited purpose of 'support[ing] a wage higher than the wage [initially] offered by the employer,' consider and seek to confirm a job applicant's wage history if such wage history was voluntarily provided."
- Hall and Krueger (2012): 1/3 of workers bargained wage before starting
- Matching outside offers

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Proposition

Consider a firm that abides by monotonicity. Fix $\epsilon > 0$, and increase a (uniformly) random worker's wage by ϵ. As $N \to \infty$, such a maneuver puts the firm in violation of monotonicty with probability $\to 1$.

Empirics

Questions

- Do firms abide by monotonicity?
- Does exogenous shock (enactment of new EPEW legislation) make firms more likely to abide by monotonicity?
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- Does exogenous shock (enactment of new EPEW legislation) make firms more likely to abide by monotonicity?
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- **To answer these questions we need a setting where wages and productivity are observable.**

Setting

- Study tenure-track and tenured faculty at US research universities
- Pay is publicly observable for most state universities by law
- Research output observable for faculty, plausibly captures most of a faculty member's productivity

Our Data

- Individual $\times$ Year panel of salary $+$ research productivity for public universities.
- Dates: $\approx 2009 - 2019$.

- Sources:
  - Publication & citation records: Card, DellaVigna, Funk & Iriberri (2023).
  - Salary & personal details: State universities, FOIA requests by media.
- Key individual covariates: University/department & gender.

Example: Mascal Pichaillat moves Brown $\rightarrow$ UCSC in 2022

- Observe: man
- In year 2021 observe:
  - 1 QJE publication in 2015 with citation counts 16,14,...,21 in years 2015,2016,...,2021, respectively
  - Works in unknown economics department
  - Rank is unknown
  - Salary is unknown
- In 2022 observe:
  - QJE publication also has 20 citations in 2020
  - Works in UCSC economics department
  - Rank is associate professor
  - Salary is X
- In 2023 observe:
  - QJE publication also has 12 citations in 2023
  - Works in UCSC economics department
  - Rank is now full professor
  - Salary is Y

Estimating Researcher Productivity

We have two broad approaches.

1) Use raw publication, journals & citation counts as several different measures of productivity (without combining them).

2) Combine publications, journal, and citations data into a single omnibus measure based.
   - We present two possible "omnibus methods" next.

Goal: Proof of concept of ideas in theory section.

- We likely cannot perfectly estimate "productivity."

Omnibus Method #1: Pope & Snydor-like approach

Based on: Pope & Snydor (AEJ, 2011)

Estimate regression of salary as a function of observables, in particular:

- Research productivity observables (pubs +cites).
- Demographic variable (gender).

To estimate a researcher's productivity, use coefficients as if everyone were the same gender.

Necessary to include gender in the regression to make sure other coefficients are set correctly.

Now we have set of {wage, imputed productivity} pairs.

Omnibus Method #1: Where do we go from here?

- Fit isotonic regression to department-year-rank(?) cells. $MSE > 0$ within cell $\Rightarrow$ failure of monotonicity.
- Does MSE change around enactment of new equal pay? E.g. California enacts new equal pay law in 2016, how does this affect UCSC.
- (Possibly related to above) Do new arrivals/departures upturn/reset the monotonicity apple cart?

Omnibus Method #2: Rationalizing Salaries

- Inspiration: Rambachan (2022, "Identifying prediction mistakes in observational data").
- Paper asks: "do (linear) preferences exist that rationalize observed choices without prediction mistakes?"
- Our version: Is there a (linear) weighting of publications at different journals that would rationalize observed salaries as monotonic?

Omnibus Method #2: Rationalizing Salaries

- Linear preferences: A university's wage function as a linear combination of K research productivity metrics.

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_K X_{iK} \tag{1}$$

- A department of N professors can be represented as a system of $\binom{N}{2}$ linear inequalities (at most).

- Example: $N = 3, K = 2$. $\binom{3}{2} = 3$ inequalities (max):
  1) $Y_1 \geq Y_2 \implies \beta_1 X_{1,1} + \beta_2 X_{1,2} \geq \beta_1 X_{2,1} + \beta_2 X_{2,2}$
  2) $Y_2 \geq Y_3 \implies ...$
  3) $Y_1 \geq Y_3 \implies ...$
- This system of inequalities is a linear programming problem with an efficient analytic solution.

Omnibus Method #2: Rationalizing Salaries

- Solution to this LP is either
  1) Empty set: Wages cannot be rationalized as monotonic.
  2) Feasible region: Wages can be rationalized as monotonic, but only if the βs weights lie inside the region, aka the "monotonicity zone." Convex polyhedron (or unbounded).

- Do actual weights βs fall within this region?
- If it is empty, then no. If not empty ...
- Idea: Estimate Equation 1 as a regression, i.e.

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + ... + \beta_K X_{iK} + \epsilon$$

- Joint test: Do the estimated β coefficients (jointly) lie within the monotonicity zone?